

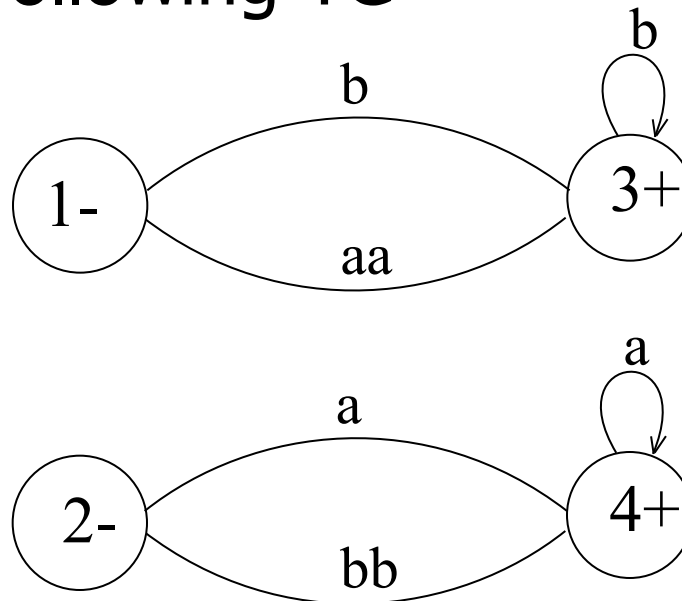
# Recap lecture 11



- Proof of Kleene's theorem part II (method with different steps), particular examples of TGs to determine corresponding REs.

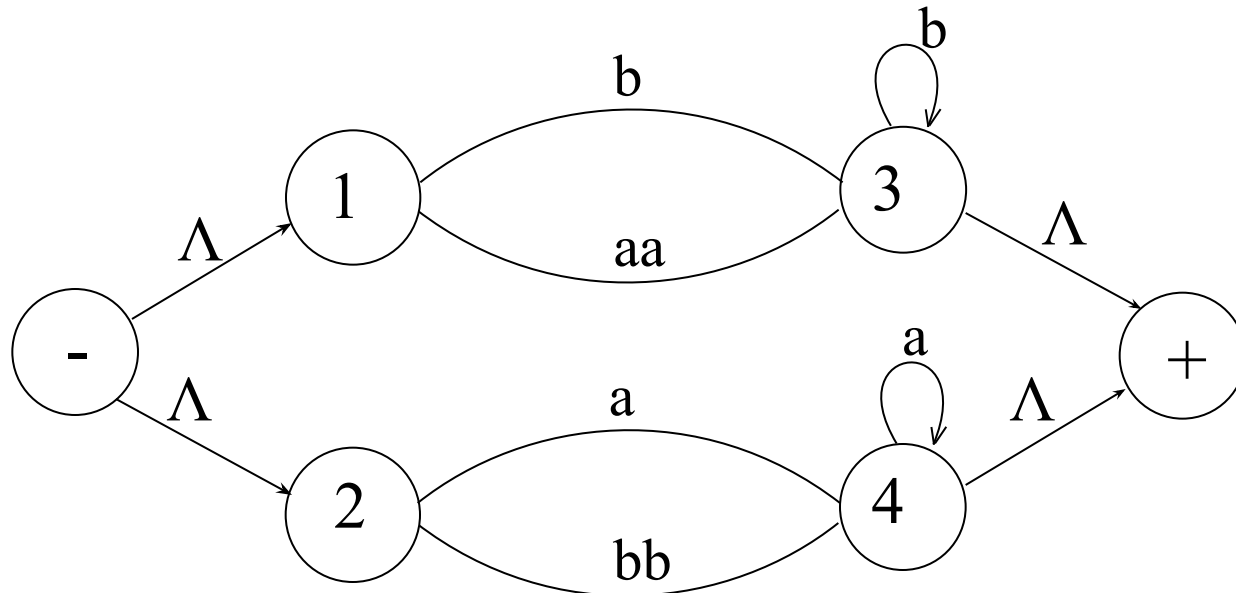
# Example

- Consider the following TG



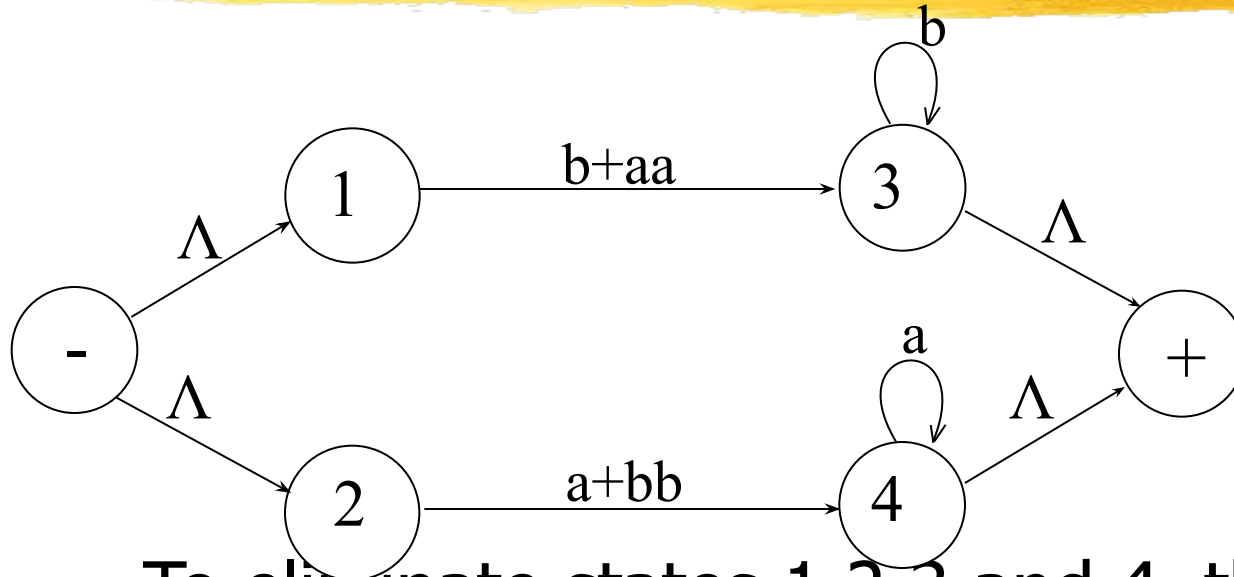
To have single initial and single final state the above TG can be reduced to the following

## Example continued ...

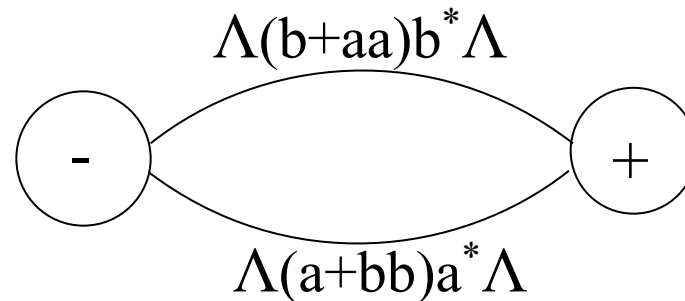


To obtain single transition edge between 1 and 3; 2 and 4, the above can be reduced to the following

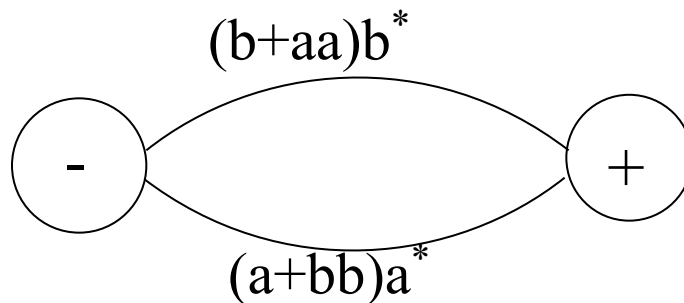
# Example continued ...



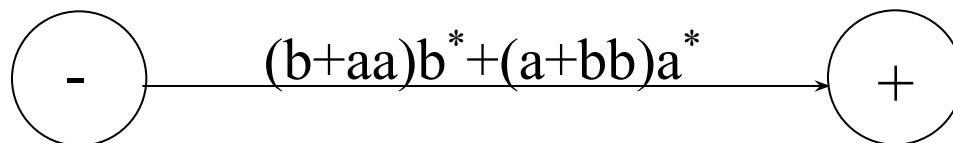
To eliminate states 1,2,3 and 4, the above TG can be reduced to the following TG



## Example continued ...



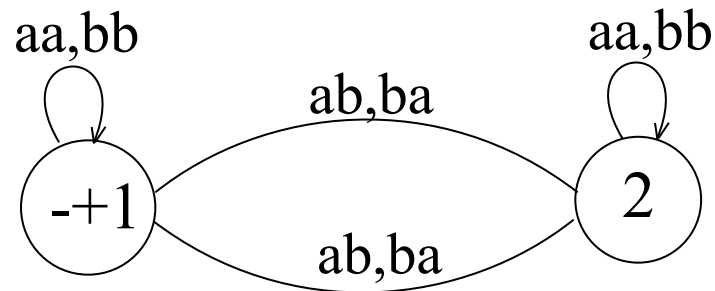
To connect the initial state with the final state by single transition edge, the above TG can be reduced to the following



Hence the required RE is  $(b+aa)b^* + (a+bb)a^*$

# Example

Consider the following TG, accepting EVEN-EVEN language

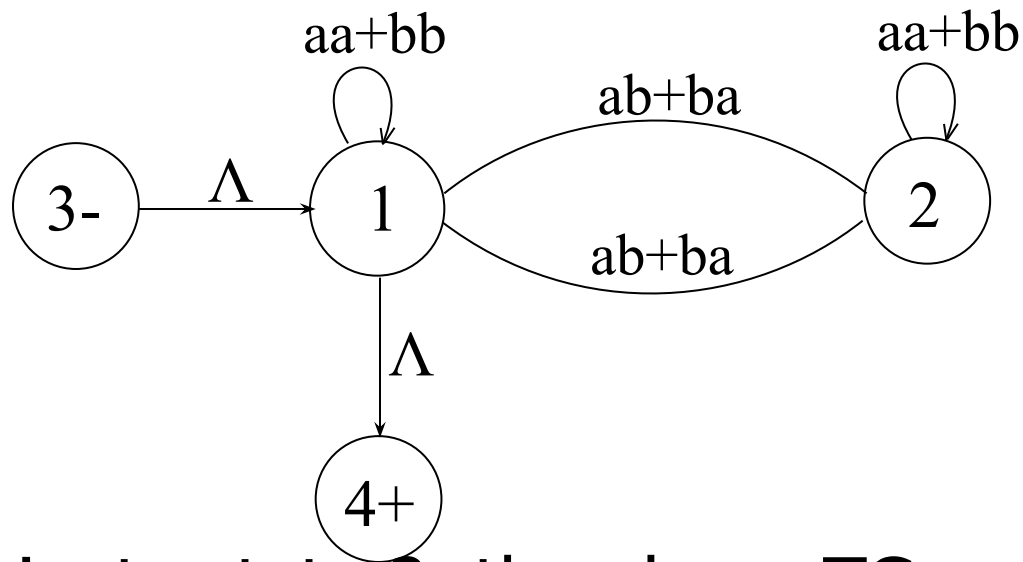


## **Example continued ...**



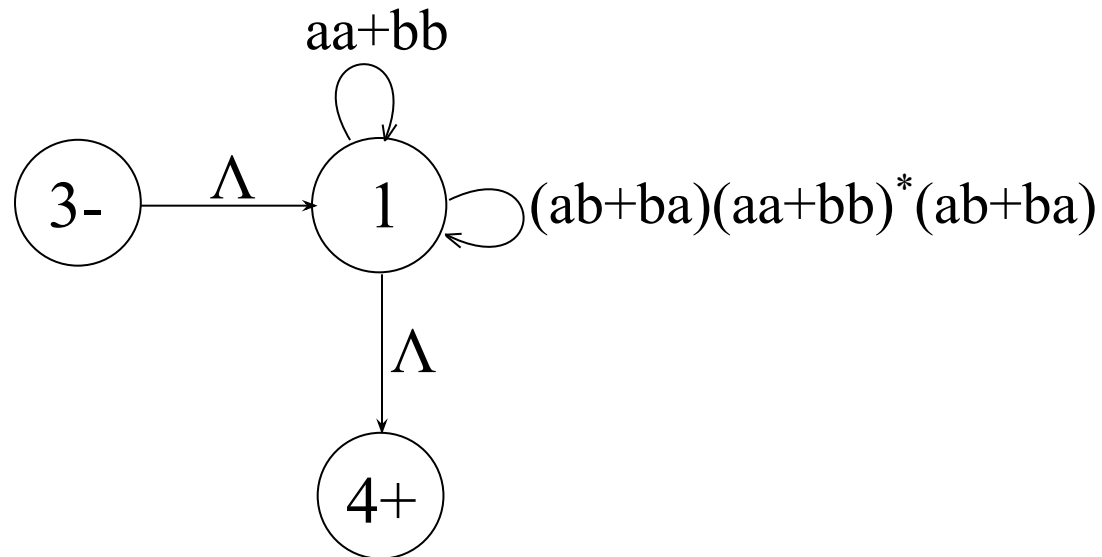
It is to be noted that since the initial state of this TG is final as well and there is no other final state, so to obtain a TG with single initial and single final state, an additional initial and a final state are introduced as shown in the following TG

## Example continued ...



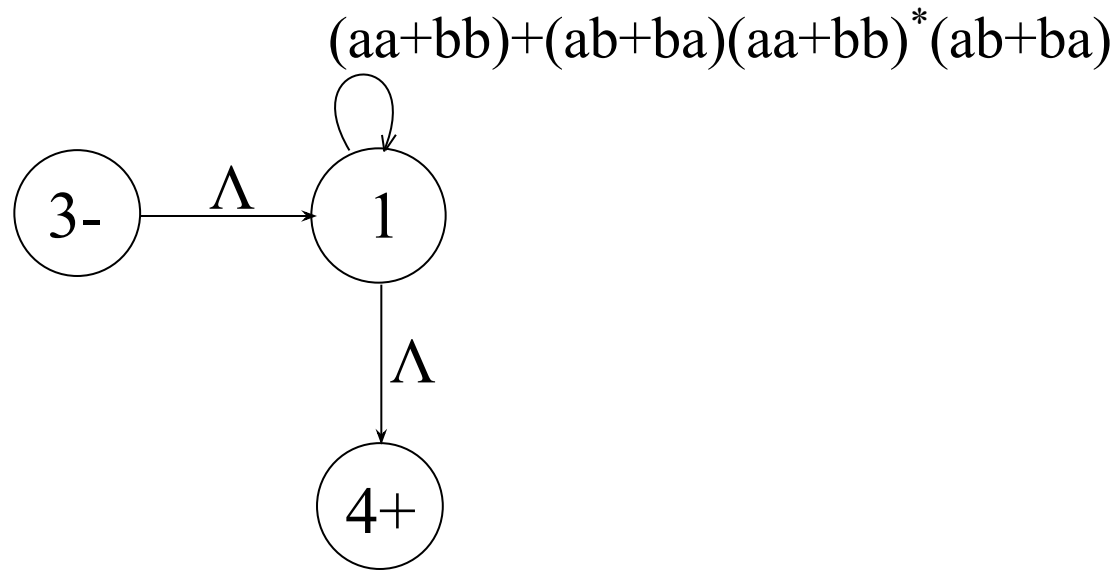
- To eliminate state 2, the above TG may be reduced to the following

## Example continued ...



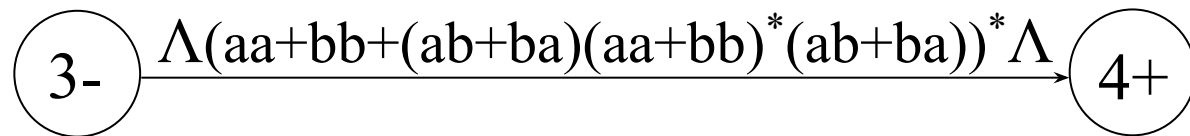
To have single loop at state 1, the above TG may be reduced to the following

## Example continued ...



To eliminate state 1, the above TG may be reduced to the following

## Example continued ...



Hence the required RE is

$(aa+bb+(ab+ba)(aa+bb)^*(ab+ba))^*$

# Kleene's Theorem Part III



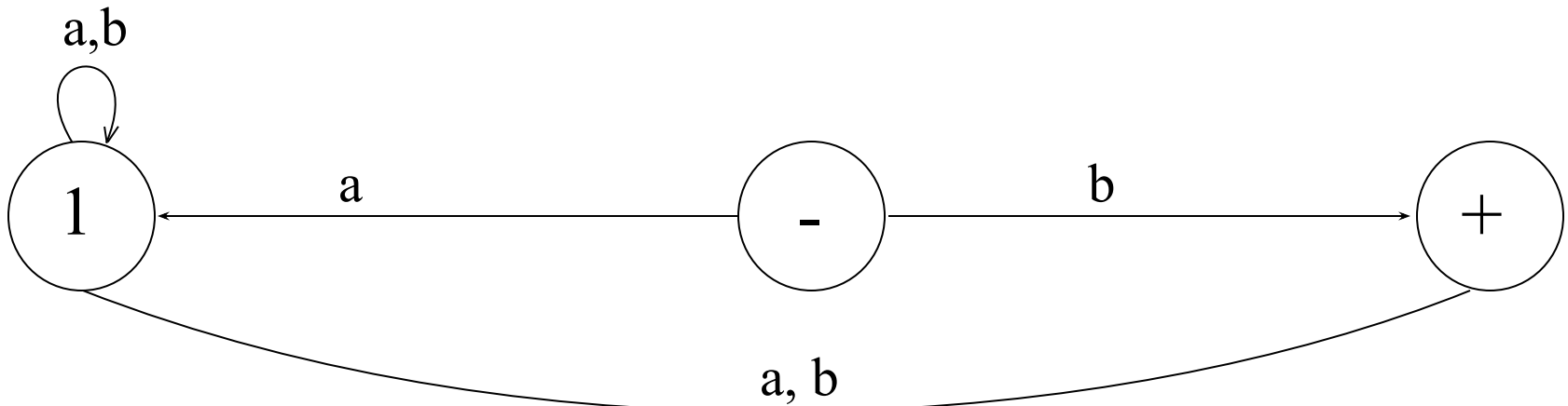
## Statement:

If the language can be expressed by a RE then there exists an FA accepting the language.

A) As the regular expression is obtained applying addition, concatenation and closure on the letters of an alphabet and the Null string, so while building the RE, sometimes, the corresponding FA may be built easily, as shown in the following examples

# Example

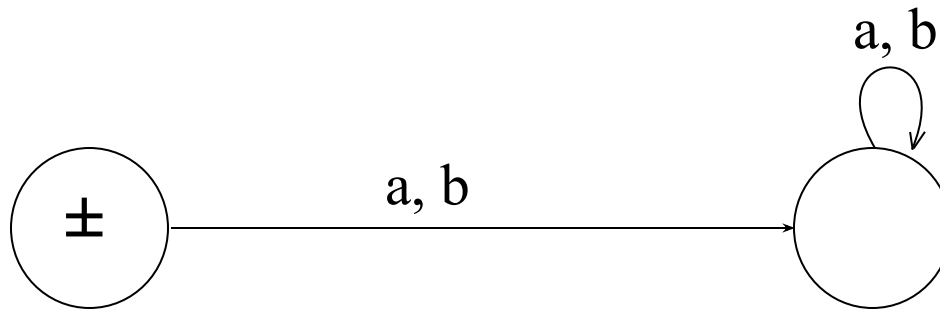
- Consider the language, defined over  $\Sigma=\{a,b\}$ , **consisting of only b**, then this language may be accepted by the following FA



which shows that this FA helps in building an FA accepting only one letter

# Example

- Consider the language, defined over  $\Sigma=\{a,b\}$ , **consisting of only**  $\square$ , then this language may be accepted by the following FA



# Kleene's Theorem Part III

## Continued ...



- B) As, if  $r_1$  and  $r_2$  are regular expressions then their sum, concatenation and closure are also regular expressions, so an FA can be built for any regular expression if the methods can be developed for building the FAs corresponding to the sum, concatenation and closure of the regular expressions along with their FAs. These three methods are explained in the following discussions

# Kleene's Theorem Part III

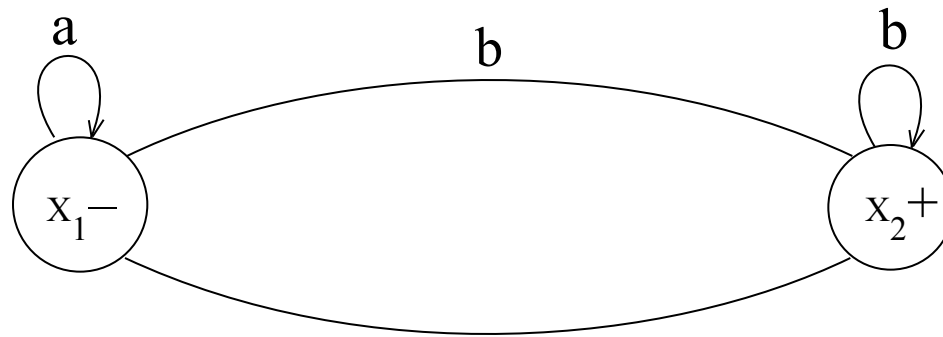
## Continued ...



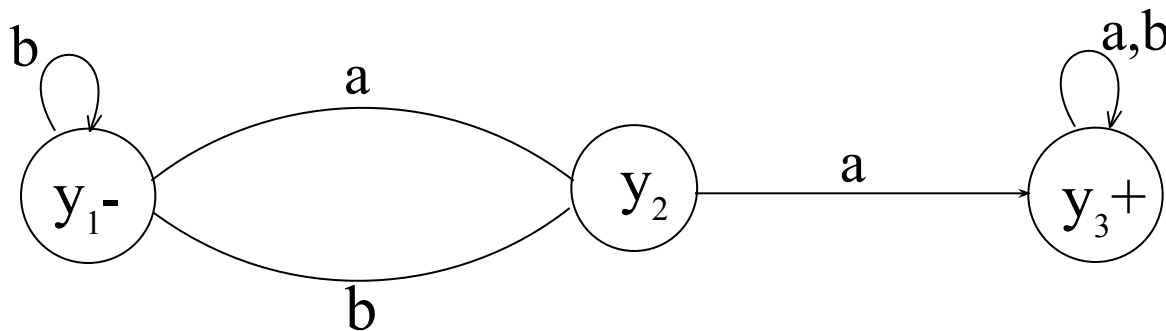
- **Method1 (Union of two FAs):** Using the FAs corresponding to  $r_1$  and  $r_2$  an FA can be built, corresponding to  $r_1 + r_2$ . This method can be developed considering the following examples

# Example

Let  $r_1 = (a+b)^*b$  defines  $L_1$  and the  $FA_1$  be



and  $r_2 = (a+b)^*aa(a+b)^*$  defines  $L_2$  and  $FA_2$  be



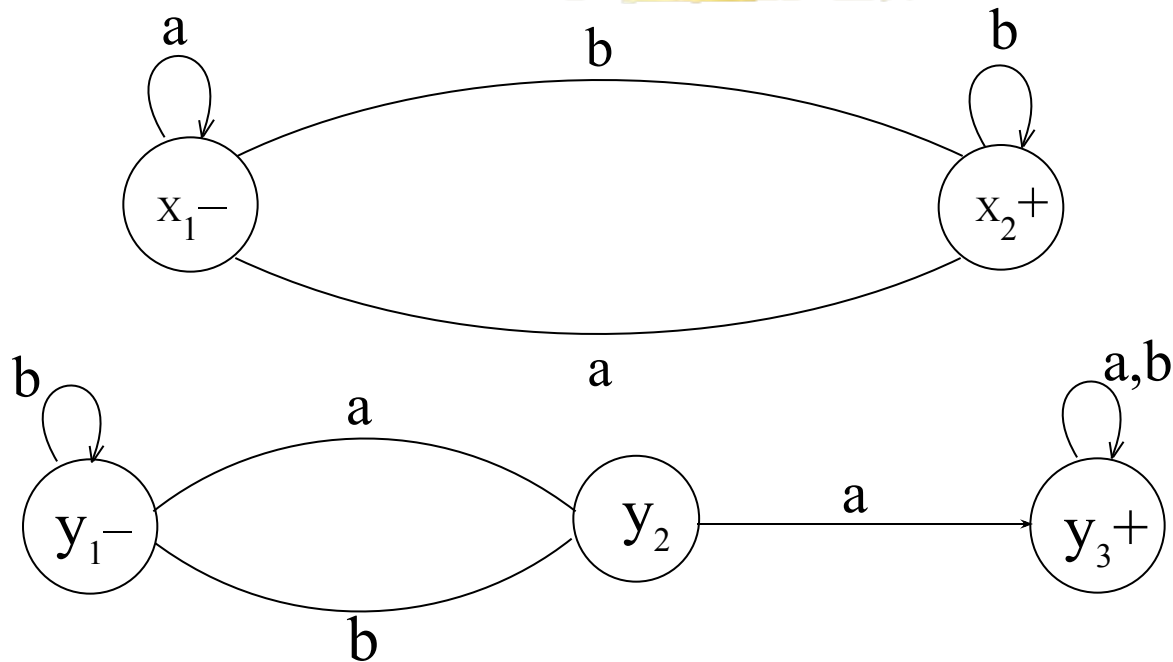
## Sum of two FAs Continued ...

Let  $FA_3$  be an FA corresponding to  $r_1 + r_2$ , then the initial state of  $FA_3$  must correspond to the initial state of  $FA_1$  or the initial state of  $FA_2$ . Since the language corresponding to  $r_1 + r_2$  is the union of corresponding languages  $L_1$  and  $L_2$ , consists of the strings belonging to  $L_1$  or  $L_2$  or both, therefore a final state of  $FA_3$  must correspond to a final state of  $FA_1$  or  $FA_2$  or both.

## Sum of two FAs Continued ...

Since, in general,  $FA_3$  will be different from both  $FA_1$  and  $FA_2$ , so the labels of the states of  $FA_3$  may be supposed to be  $z_1, z_2, z_3, \dots$ , where  $z_1$  is supposed to be the initial state. Since  $z_1$  corresponds to the states  $x_1$  or  $y_1$ , so there will be two transitions separately for each letter read at  $z_1$ . It will give two possibilities of states either  $z_1$  or different from  $z_1$ . This process may be expressed in the following transition table for all possible states of  $FA_3$ .

# Example continued ...

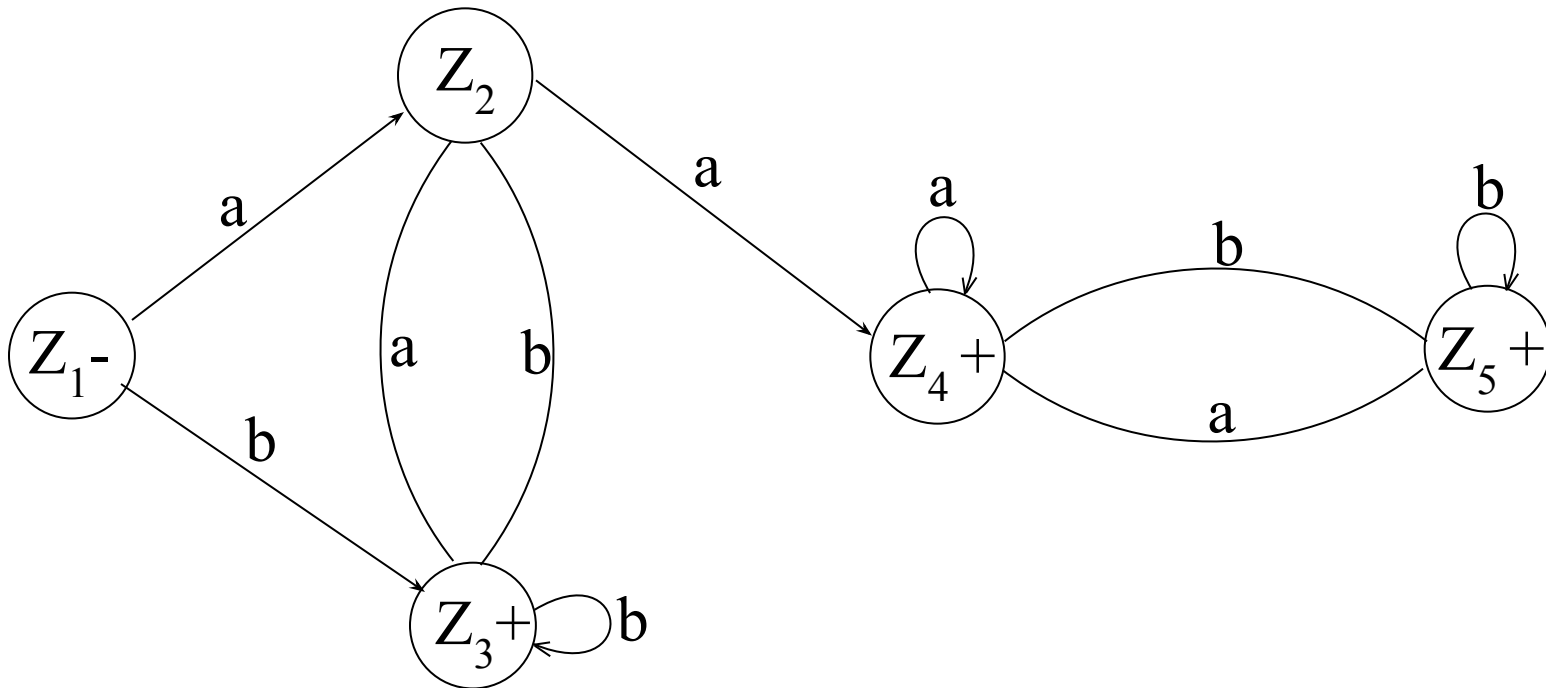


| Old states                | New states after reading |                         |
|---------------------------|--------------------------|-------------------------|
|                           | a                        | b                       |
| $z_1^- \sqcap (x_1, y_1)$ | $(x_1, y_2) \sqcap z_2$  | $(x_2, y_1) \sqcap z_3$ |

# Example continued ...

| Old States                | New States after reading |                         |
|---------------------------|--------------------------|-------------------------|
|                           | a                        | b                       |
| $z_2 \sqcap (x_1, y_2)$   | $(x_1, y_3) \sqcap z_4$  | $(x_2, y_1) \sqcap z_3$ |
| $z_3^+ \sqcap (x_2, y_1)$ | $(x_1, y_2) \sqcap z_2$  | $(x_2, y_1) \sqcap z_3$ |
| $z_4^+ \sqcap (x_1, y_3)$ | $(x_1, y_3) \sqcap z_4$  | $(x_2, y_3) \sqcap z_5$ |
| $z_5^+ \sqcap (x_2, y_3)$ | $(x_1, y_3) \sqcap z_4$  | $(x_2, y_3) \sqcap z_5$ |

## Example continued ...



RE corresponding to the above FA may be  
$$r_1 + r_2 = (a+b)^*b + (a+b)^*aa(a+b)^*$$

# Summing Up



- Examples of writing REs to the corresponding TGs, RE corresponding to TG accepting EVEN-EVEN language, Kleene's theorem part III (method 1: union of FAs), examples of FAs corresponding to simple REs, example of Kleene's theorem part III (method 1) continued